

Sound — Workout

ICSE Class 7 Physics · 10 MCQs · 1 mark each · Name: _____ Date: _____

Syllabus: Whole chapter: production and propagation of sound, wave terms and $v = f \lambda$, classification by frequency, characteristics, echoes and SONAR.

Instructions: Each question has one correct option. Tick or circle the letter of the best answer. Attempt all questions.

1. Sound is produced by vibrations and needs a medium; it travels fastest through:

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|------------|-----------|
| A. liquids | B. solids |
| C. vacuum | D. gases |

2. A sound wave with widely spaced rarefactions and few vibrations per second has a:

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|----------------|---------------------|
| A. high pitch | B. short wavelength |
| C. large speed | D. low frequency |

3. A tuning fork of frequency 425 Hz sounds in air ($v = 340$ m/s). Its wavelength is:

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|-------------|-----------|
| A. 144500 m | B. 2 m |
| C. 0.8 m | D. 1.25 m |

4. The silent whistles used to call dogs usually produce sound that is:

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|--------------------|---------------|
| A. no sound at all | B. ultrasonic |
| C. infrasonic | D. audible |

5. Beating a drum harder, without changing anything else, increases the sound's:

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|--------------|-------------|
| A. pitch | B. loudness |
| C. frequency | D. speed |

6. Tightening a guitar string raises its frequency, so the note becomes:

- | | |
|-----------|--------------------|
| A. slower | B. softer |
| C. louder | D. higher in pitch |

7. Thunder is heard 4 s after the lightning flash. Taking the speed of sound as 340 m/s, the lightning struck about:

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|----------------|---------------|
| A. 1360 m away | B. 340 m away |
| C. 680 m away | D. 85 m away |

8. Ultrasound sent into the sea by a ship returns in 2 s (speed in water 1500 m/s). The sea is ____ deep.

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|-----------|-----------|
| A. 375 m | B. 750 m |
| C. 1500 m | D. 3000 m |

9. Echoes are not heard in an ordinary small room because the walls are:

- | | |
|-------------------------------------|-------------|
| A. too close (less than about 17 m) | B. too soft |
| C. too hard | D. too cold |

10. A stethoscope carries the sound of the heartbeat to the doctor's ears mainly by:

- | | |
|--|---------------|
| A. repeated reflection of sound inside its tubes | B. absorption |
| C. refraction | D. radiation |

Answer Key

1. B

2. D

3. C

4. B

5. B

6. D

7. A

8. C

9. A

10. A